Novel Volumetric Sub-region Segmentation in Brain Tumors

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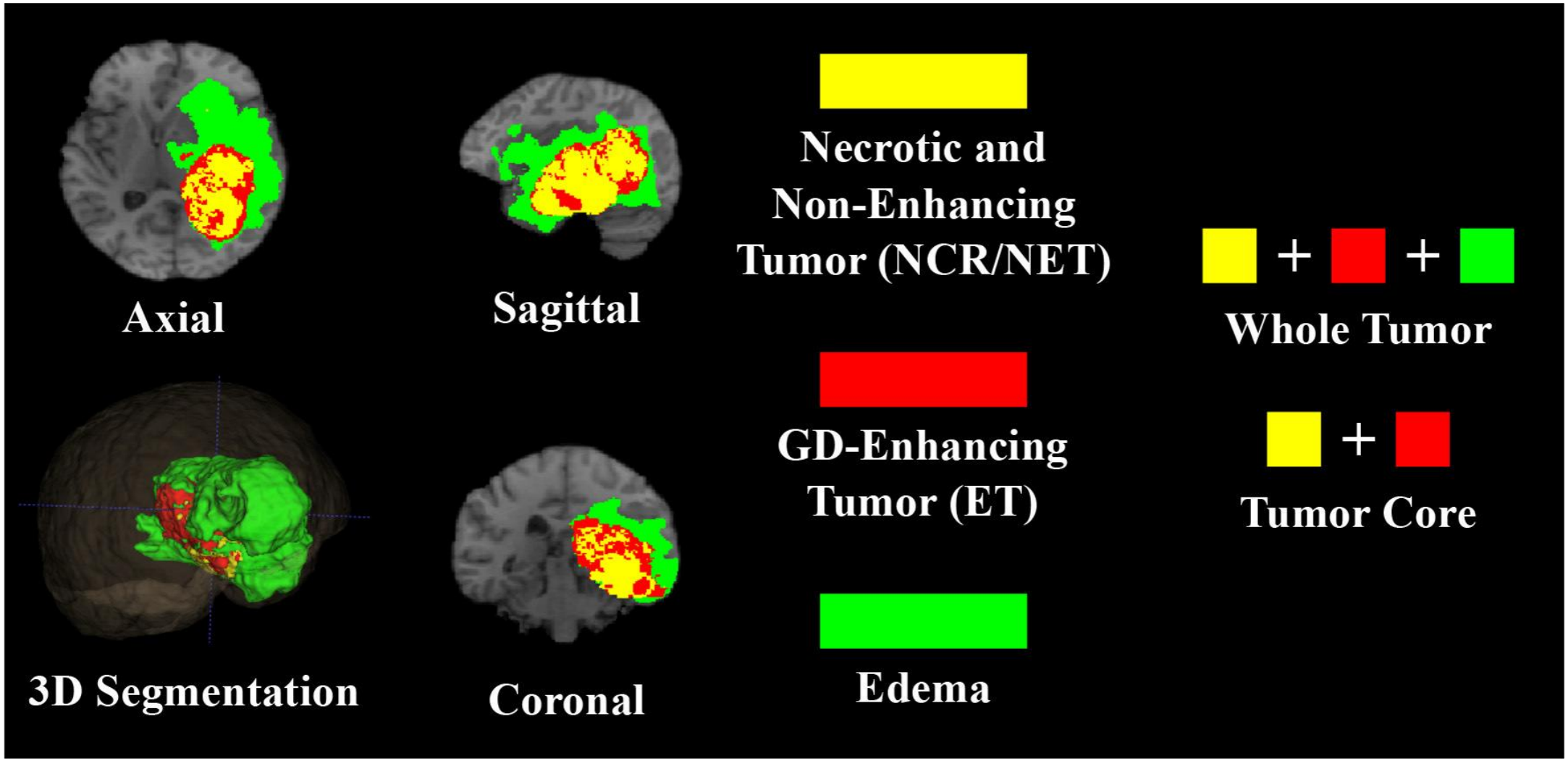
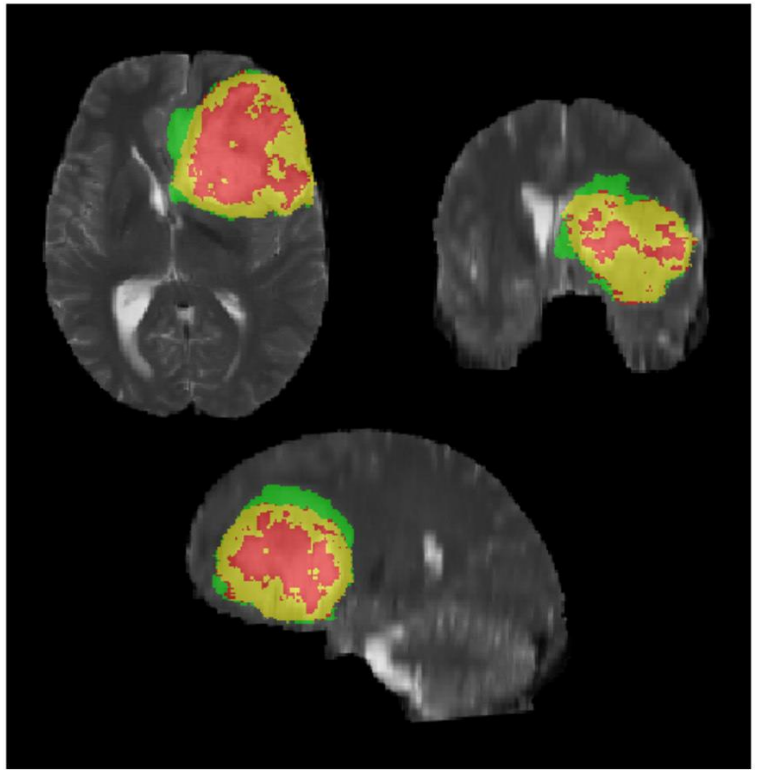
frontiers
in Computational Neuroscience

2020.3.5
胡敬玉

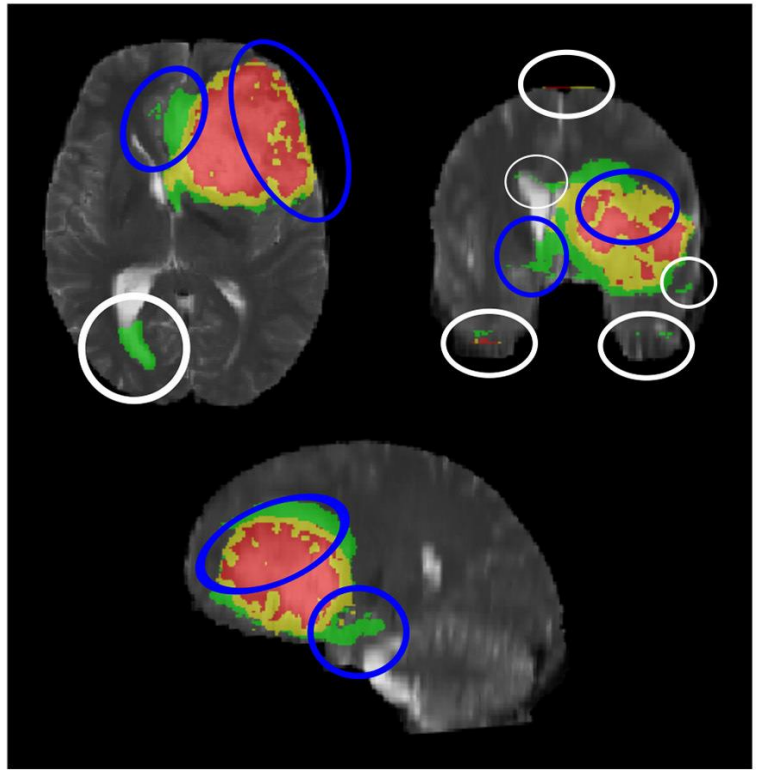
问题


Axial
Sagittal
3D Segmentation
Coronal
Necrotic and
Non-Enhancing
Tumor (NCR/NET)
GD-Enhancing
Tumor (ET)
Edema
+ +
Whole Tumor
+
Tumor Core

问题



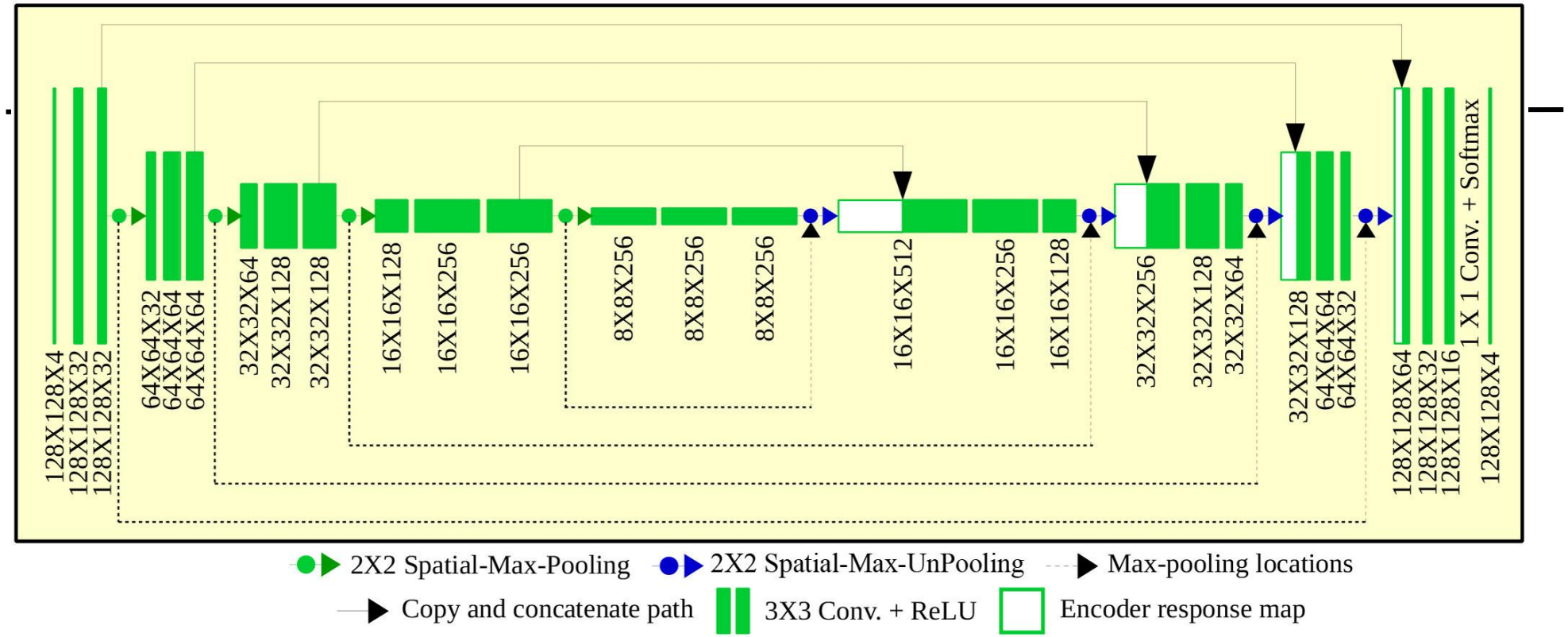
Ground-truth



Predicted-segmentation

- Error around boundary
- False positive at the top and bottom ends of the MRI volume

A



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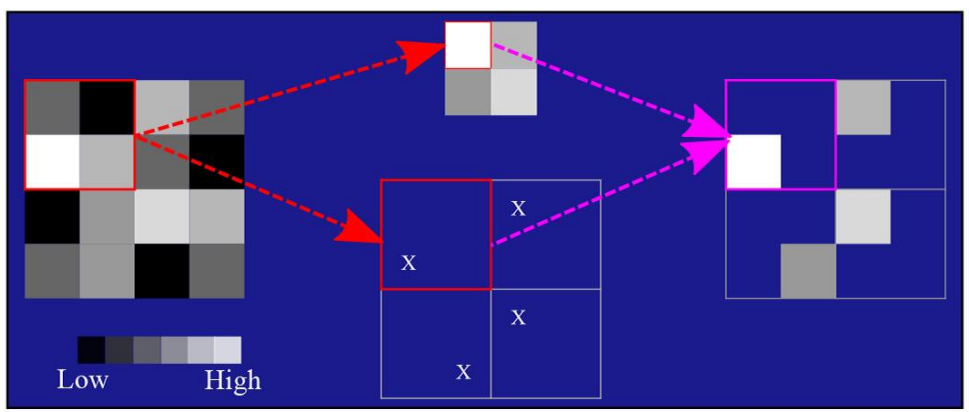


2X2 Spatial-Max-Pooling and UnPooling

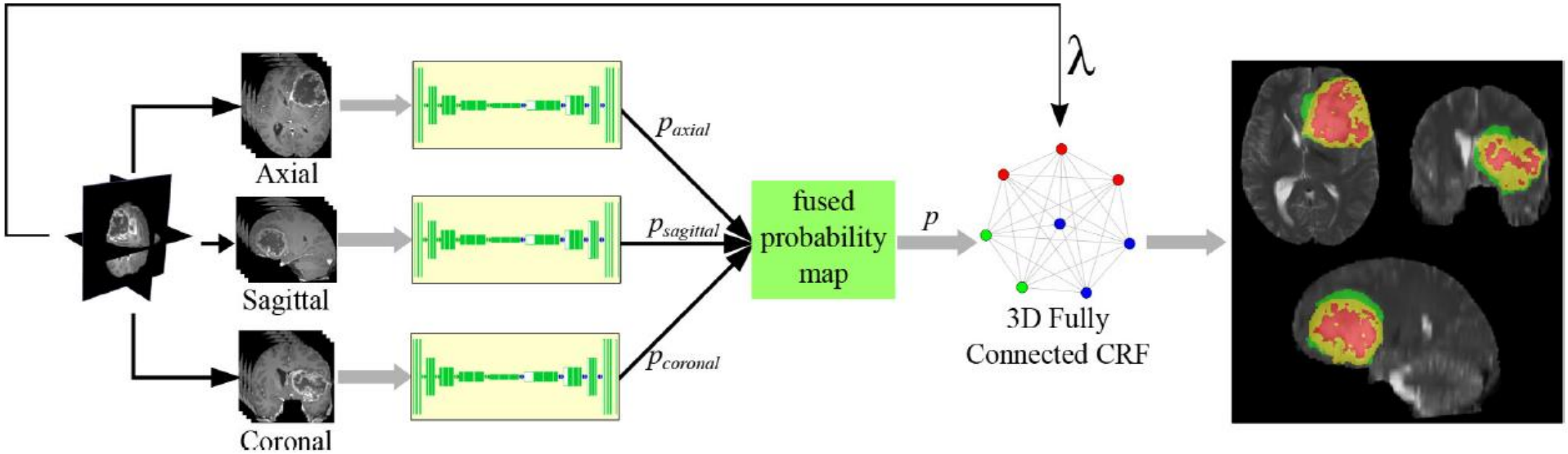
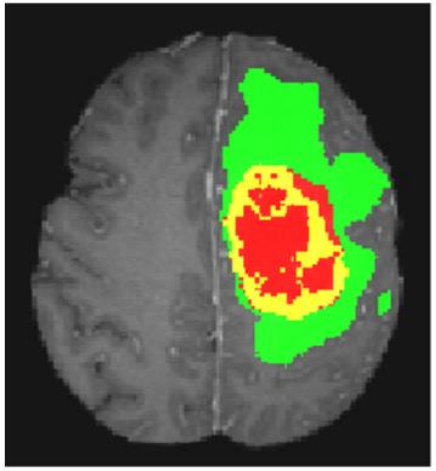


FIGURE 4 | Aggregated architecture combining multiple planes, with CRF-based refinement.



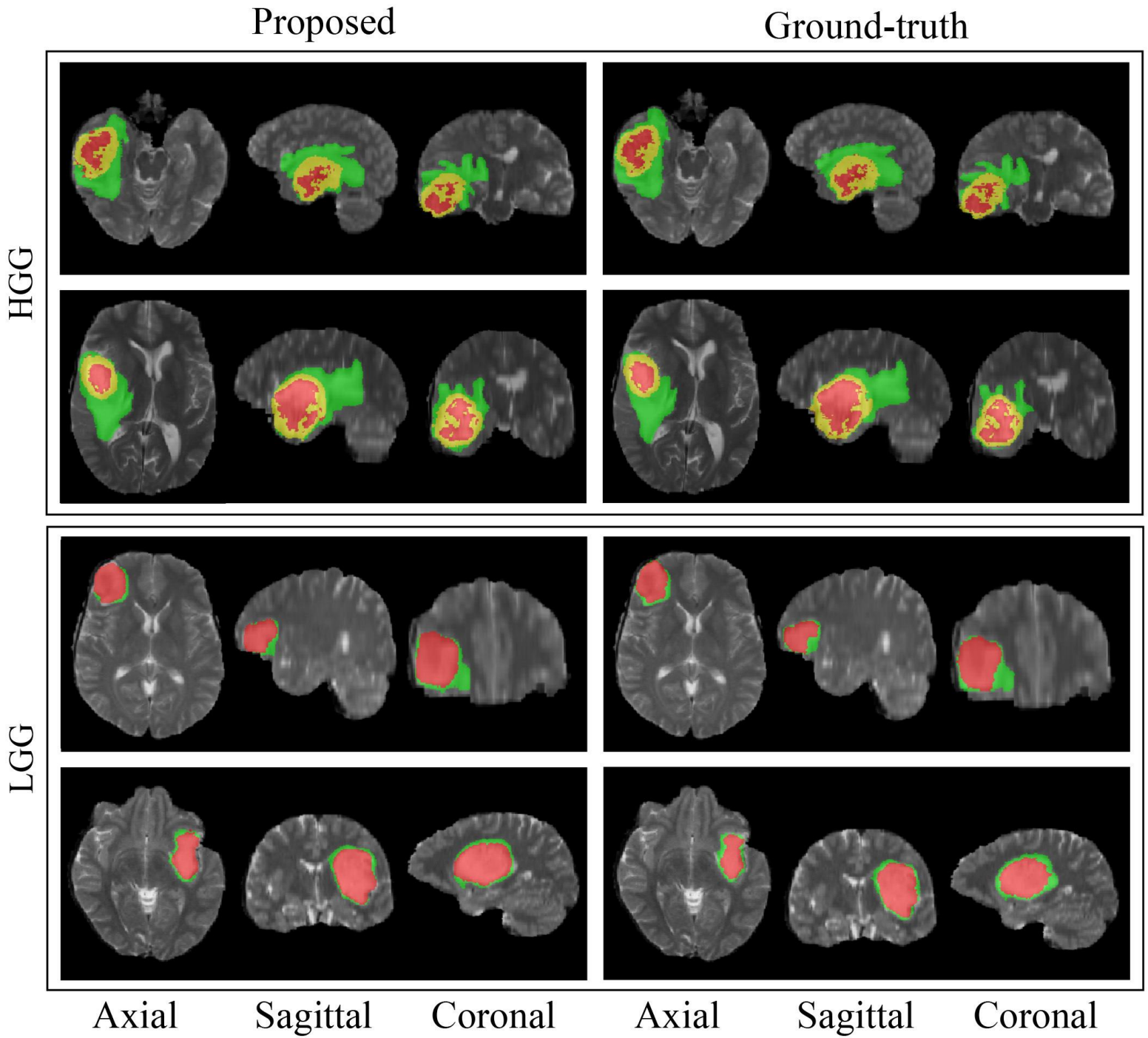
ED
NCR/NET
ET

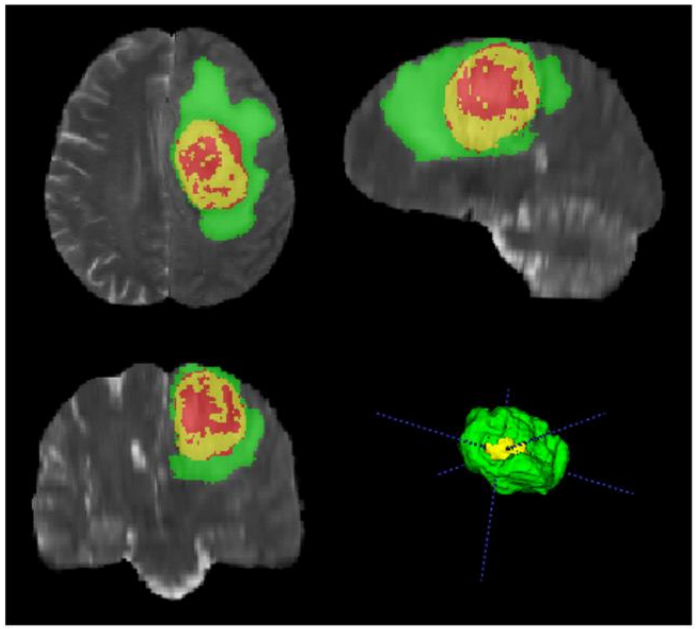
Class	Voxel Count	%
Background	8790699	98.46
ED	90806	1.02
ET	26264	0.29
NCR/NET	20231	0.23

$$WGDL = 1 - 2\frac{\sum_{c=1}^{|C|} wa_c \sum_{n=1}^{N} G_{cn}P_{cn}}{\sum_{c=1}^{|C|} wa_c \sum_{n=1}^{N} G_{cn} + P_{cn}},$$

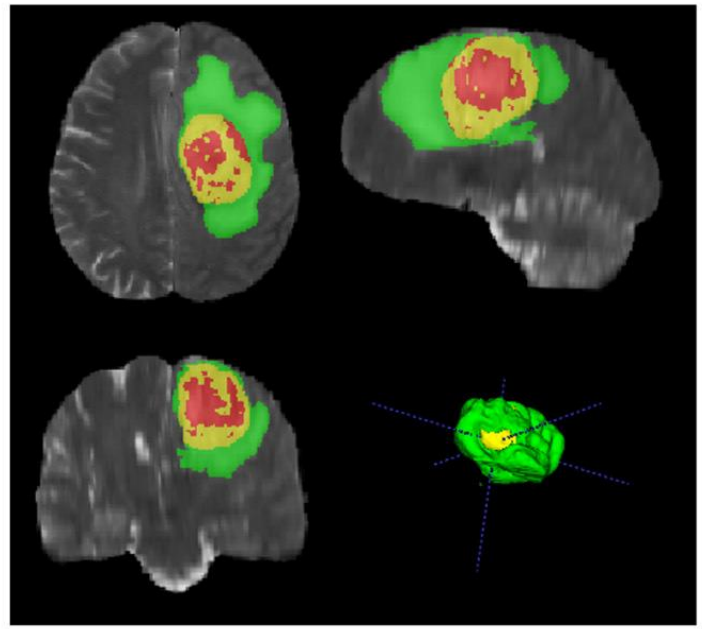
$$wa_c = \frac{1}{(\sum_{n=1}^{N} G_{cn})^2},$$

$$WLL = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{|C|} ws_c G_{cn} \log(P_{cn}),$$

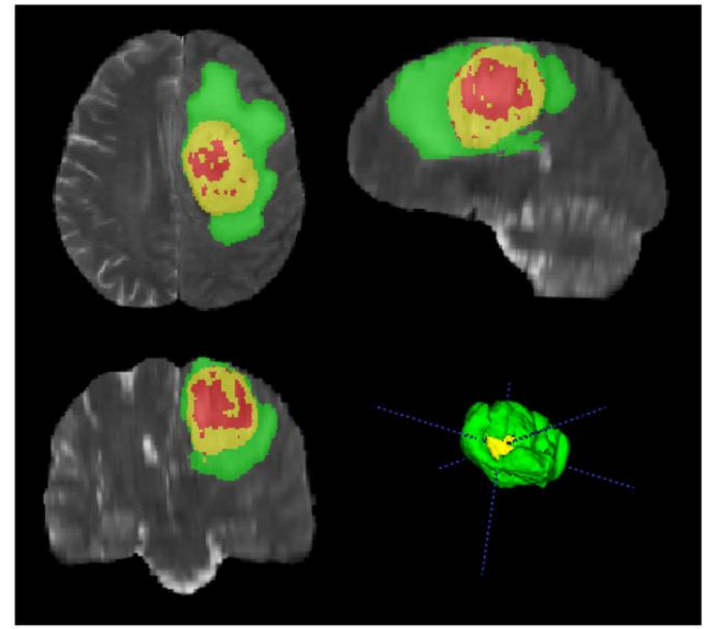

Proposed
Ground-truth
HGG
LGG
Axial
Sagittal
Coronal
Axial
Sagittal
Coronal



Ground truth



MPS-CNN



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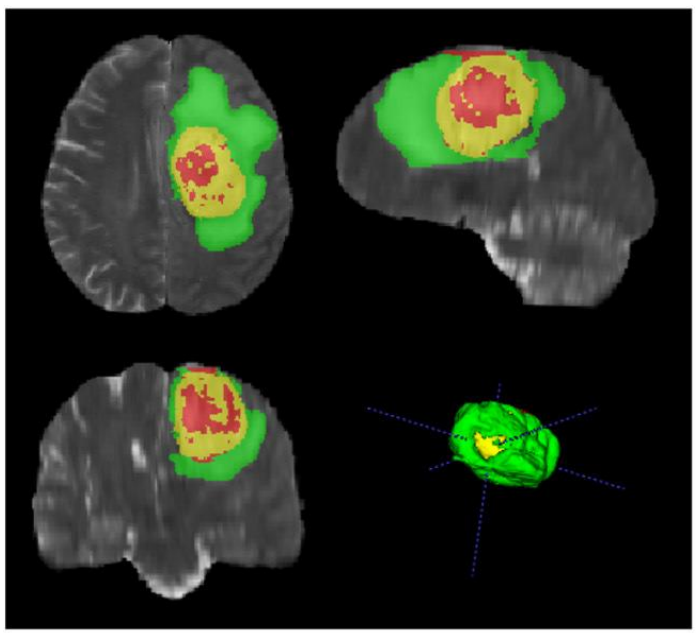

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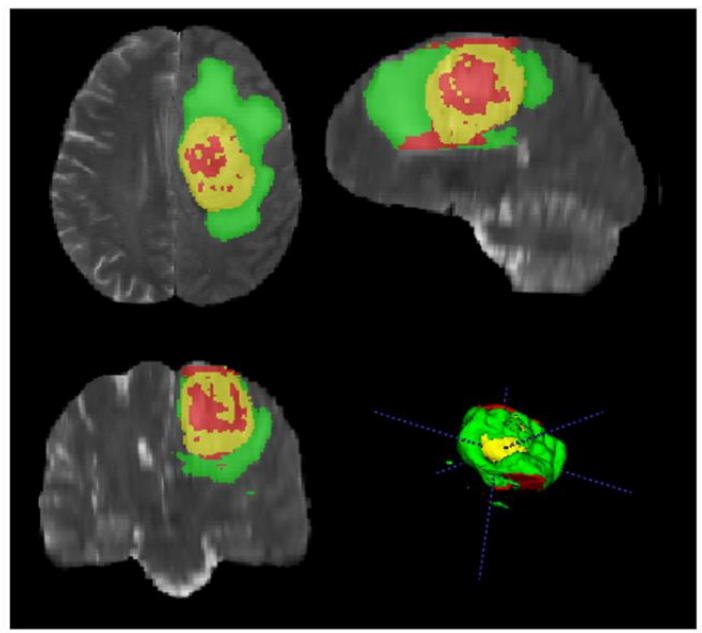
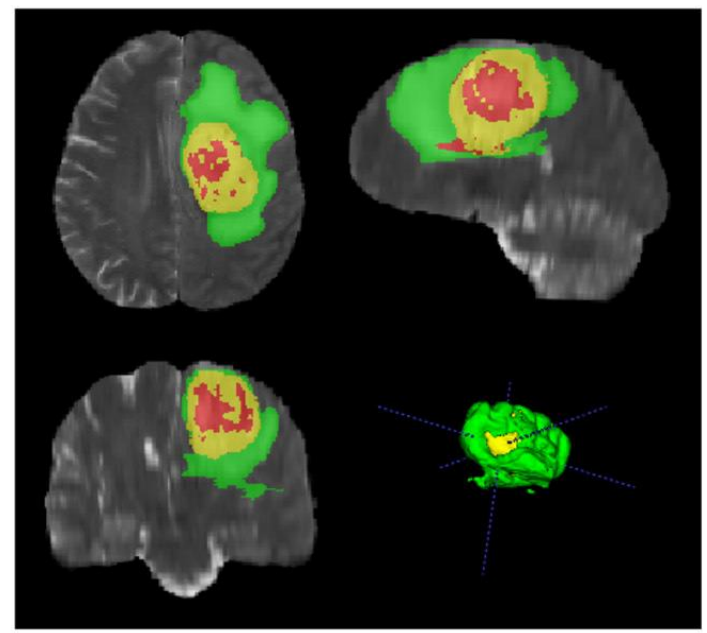


C



D

Thank You!